A Relational Database for Brazilian Jiu-Jitsu Match Tracking

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***Abstract*—This report documents the design and implementation of a relational database that tracks Brazilian Jiu-Jitsu fighters, the gyms they train at, and the results of their matches. The database was designed using five entities and normalised to remove redundancy, then built in MySQL using phpMyAdmin and filled with sample data that was imported from CSV files. Six SQL queries were then written to test the database, and one part of the scenario that would be difficult to implement in SQL is discussed.**

I. Introduction

Brazilian Jiu-Jitsu(BJJ) is a grappling martial art which consists of 1 vs 1 matches between fighters, in typical competitions fighters can win their matches via submissions, points, and various other methods. This means tracking fighter's gyms and results in a single spreadsheet can quickly lead to repeated data and inconsistent records.Therefore, the aim of this project is to tackle this problem by managing scenarios as a relational database that can answer questions such as who won a match,their method of victory, and how often each fighter wins.

The database consists of five entities: gym, fighter, VicotryType, match, FighterMatchPerformance.This is because a fighter takes part in many matches which involve two fighters, FighterMatchPerfomance resolves this many-to-many relationship by using a composite primary key that is made of both FighterID and MatchID.Win rate is a derived value that is calculated by a query rather than being stored, this avoids redundancy and out of date values.The database design assumes that each match has two performance records,meaning that each fighter belongs to one gym, each match has one victory method, and all data are fictional.Section II will present the design, Section III the implementation, Section IV the queries, and Section V will discuss the complexity.

II. Database Design

A. Entities and Relationships

B. Normalisation

TABLE I
Normal form of each entity

Entity	NF	Justification
Gym		
Fighter		
VictoryType		
Match		
FighterMatch-Performance		

C. Data Types and Integrity

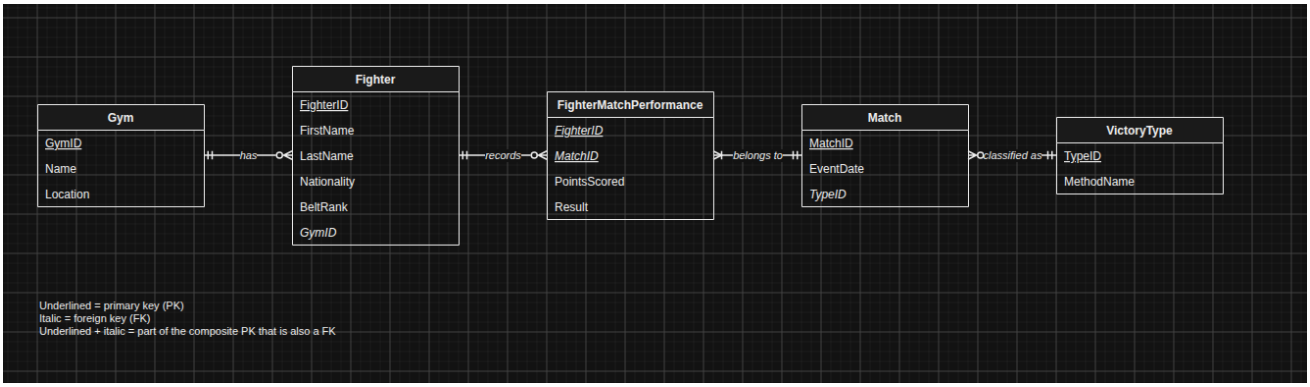


Fig. 1. Entity-relationship diagram. Underlined attributes are primary keys, italic attributes are foreign keys, and underlined italic attributes form the composite primary key of FighterMatchPerformance.